GHOST PROTOCOL — SOC WORKSTATION DASHBOARD DESIGN SPECIFICATION

Engineering Specification for Antigravity | Three-Part Cyber-Intelligence Layout

1. ARCHITECTURAL OVERVIEW & LAYOUT SPECIFICATION

The GHOST Protocol Workstation is an industrial, cyber-threat intelligence dashboard designed specifically for SOC analysts and threat researchers. Drawing aesthetic and structural inspiration from high-density military/intelligence interfaces (dark palette, zero animation, strict grid alignment, high data density), the UI is divided into a mandatory **Three-Part Layout**:

Panel Section	Width / Flex	Core Responsibility & UI Componentry
LEFT SIDEBAR (Navigation & Modules)	240px Fixed	Global module selector featuring 17 distinct functional sections. Houses system status badge, active environment indicator (Docker Lab vs. Tor/Whonix), and primary view switcher.
CENTER WORKSPACE (Dynamic View Port)	Flex 1 (Rest)	Context-aware primary workspace that dynamically re-renders based on the active sidebar selection. Displays threat feeds, interactive STIX 2.1 property graphs, entity resolution matrices, XAI cards, and GHOST AI investigation console.
RIGHT TELEMETRY (Live Operations)	320px Fixed	Real-time operational telemetry HUD. Displays GHOST autonomous agent state, Whonix SOCKS5 connection state, Playwright session vault health, ML model execution status, live risk score gauge, STIX/MITRE counters, and fail-closed circuit status.

2. DESIGN SYSTEM & VISUAL DIRECTIVE

Antigravity must adhere strictly to the following dark cyber-intelligence design tokens. No CSS/JS animations or transition lags are permitted—all state updates must render instantly.

Design Token	Hex Value	Usage & Application
Background Base	#0b0f19	Primary application dark background base
Card / Surface	#111827	Container cards, panel backgrounds, modal dialogs
Border / Rule	#1e293b	1px subtle grid borders, divider lines, card outlines
Accent Cyber Blue	#2563eb	Primary action buttons, active navigation states, graph nodes
Threat Red (High)	#dc2626	High Risk Scores (>= 0.70), critical CVEs, ransomware claims
Alert Amber (Med)	#d97706	Moderate Risk (0.40 - 0.69), 503 barriers, unverified aliases
Status Green (Safe)	#059669	Whonix SOCKS5 connected, session active, ML pipeline healthy
Monospace Font	JetBrains Mono / Fira	Hashes, IP addresses, CVE IDs, SOCKS5 proxies, code blocks

3. LEFT SIDEBAR MENU SPECIFICATION (17 MODULES)

The left sidebar houses the complete functional menu. Antigravity must wire each menu item to its corresponding REST API endpoint and center workspace view component:

Menu Item	Icon / Key	Target Workspace Component & Real Data Binding
1. Dashboard	grid	Executive overview: Threat map, recent alert cards, risk distribution, agent status summary.
2. Source Discovery	compass	Discovery queue (`pending_discovery_queue`), Daunt.link mirrors, Ahmia search API, deepdarkCTI feeds.
3. Sources	database	Master source registry (`sources_registry`): Status CRUD (Active/Deactivated), reachability pings.
4. Live Observation	activity	Real-time stream feed of incoming raw/clean HTML, SHA-256 hashes, MinHash LSH deduplication state.
5. Threats	alert-triangle	Filtered CTI threat stream (Exploit types, RCEs, credential dumps, database leaks, access sales).
6. Entities	users	Canonical Threat Actor profile directory, handles, aliases, first/last seen timestamps, co-occurrences.
7. Correlation	git-merge	Composite Entity Resolution Engine (Jaro-Winkler + DistilBERT Cosine + Graph Jaccard) candidate links.
8. Infrastructure	server	Monitored `.onion` hosts, IP addresses, domains, crypto wallets, and host migration history (`REPLACED_BY`).
9. CTI	shield	Structured IoC database (CVEs, file hashes, Bitcoin/Monero wallets, malware families).
10. STIX	share-2	OASIS STIX 2.1 Graph Visualizer & JSON Bundle Exporter (SDOs, SCOs, SROs).
11. MITRE	layers	MITRE ATT&CK; Enterprise Matrix heat map (Mapped T-codes: T1190, T1486, mitigations).
12. XAI	cpu	Two-Tier Explainable AI Console: Tier 1 Token Highlights + Tier 2 SHAP/LIME Feature Attribution Bar Charts.
13. OSINT	search	On-Demand Sherlock Username Handle Lookup Wrapper (Transient surface web queries, zero DB storage).
14. Investigations	briefcase	Case management workspace for saving, annotating, and exporting analyst investigation reports.
15. Timelines	clock	Longitudinal activity timeline showing entity evolution, post sequences, and infrastructure shifts.
16. GHOST AI	terminal	Natural Language Query Console ("Show all LockBit victims with RCE exploits in 2026").
17. Settings	settings	System configuration: Whonix SOCKS5 proxy port, Playwright browser limits, risk alert threshold (0.70).

4. RIGHT TELEMETRY PANEL SPECIFICATION (HUD)

The right panel provides continuous real-time operational feedback. Antigravity must implement live SSE/WebSocket telemetry data bindings for the following 9 HUD modules:

Telemetry HUD Module	Displayed Metrics & Data Fields	State Indicator / Alert Rule
1. GHOST Agent State	Status: RUNNING / IDLE / PAUSED; Active Worker Threads; Current Task ID.	Green pulse dot for active execution.
2. Whonix / Tor Boundary	Proxy: socks5h://10.152.152.10:9050; IP Leak Guard: ACTIVE; Circuit Latency: 340ms.	RED alert if socket check fails (Fail-closed).
3. Browser / Session Vault	Engine: Playwright SOCKS5; Active Cookies: dread_session; Volatile Vault Health: 100%.	Amber warning if session cookie expires.
4. Observation Status	State: CONTENT_ACCESSIBLE (200) / 503_BARRIER; Last Crawl: 2s ago; Pages/Min: 14.	State badge color matches HTTP response.
5. AI Pipeline Health	DistilBERT: LOADED (CUDA); XGBoost: READY; Temporal AE: READY; HM-TIF: ACTIVE.	Green status check across all 4 models.
6. Live Risk Score Gauge	Fused Dynamic Risk Score: 0.88 / 1.00; Modality Weights: Text (0.45), Net (0.35), Temp (0.20).	Red numerical gauge if Risk >= 0.70.

7. CTI / STIX Counters	Total IoCs: 1,420; Active CVEs: 84; Generated STIX Bundles: 312.	Real-time counter increment animations.
8. MITRE T-Code Summary	Top Mapped T-Codes: T1190 (Exploit Public-Facing), T1486 (Data Encrypted), T1078.	Clickable T-code pills linking to Matrix.
9. System Health & Storage	CPU: 12%; RAM: 3.4GB / 16GB; SQLite DB: darkwatch.db (42MB); Disk Free: 128GB.	Storage health bar indicator.

5. CENTER WORKSPACE COMPONENT SPECIFICATION

Antigravity must construct the center workspace using 4 core UI component blueprints:

A. Explained Alert Card Component (Threats / Dashboard View)

Displays a processed threat observation. Must render: (1) Header with Source URL, Timestamp, and Dynamic Risk Score Badge; (2) Preprocessed Text Payload with Tier 1 DistilBERT Cross-Attention Token Highlights; (3) Mapped MITRE ATT&CK; Pill Tags (`T1190`, `T1486`); (4) Tier 2 SHAP Feature Attribution Bar Chart; (5) Export STIX 2.1 JSON Button; (6) On-Demand Sherlock Handle Lookup Button.

B. Interactive STIX 2.1 Graph Visualizer (STIX / Correlation View)

A force-directed graph canvas (using Cytoscape.js or D3.js). Renders STIX Domain Objects (SDOs: Threat Actor, Malware, Vulnerability, Identity) as colored node icons, and STIX Relationship Objects (SROs: `uses`, `targets`, `indicates`, `replaced_by`) as directed edges labeled with confidence scores.

C. Entity Profile & Alias Resolution Matrix (Entities / Correlation View)

Displays canonical threat actor profiles. Shows: Handle, Known Aliases, First-Seen/Last-Seen UTC, Associated Crypto Wallets, Malware Tools, and the Composite Resolution Match Breakdown (Jaro-Winkler + DistilBERT Cosine Vector + Graph Jaccard). Allows analyst manual merge confirmation.

D. GHOST AI Natural Language Console (GHOST AI View)

An interactive CTI query interface. Accepts natural language queries (e.g., *'Find all LockBit ransomware claims targeting healthcare with RCE exploits'*), translates them into structured SQL/Graph queries against `darkwatch.db`, and renders matching evidence cards with full provenance lineage.

6. IMPLEMENTATION ROADMAP & ACCEPTANCE CRITERIA

Phase 1: Dashboard Layout Shell & REST API Routing

Implement CSS Grid template for the Three-Part Layout (240px sidebar, flex center, 320px telemetry HUD). Build Flask/Streamlit routing endpoints for all 17 sidebar views.

Phase 2: Live Telemetry WebSocket / SSE Service

Wire `darkwatch/dashboard/telemetry_service.py` to stream real-time JSON metrics (Whonix socket health, Risk score, STIX counts, agent status) to the right HUD panel.

Phase 3: Center Workspace View Components

Build Explained Alert Cards, Cytoscape STIX Graph Canvas, Entity Resolution Matrix, and GHOST AI Natural Language Console.

Phase 4: Integration Verification

Run end-to-end integration test (`test_dashboard_e2e.py`) verifying that crawling a lab endpoint updates the live telemetry HUD, creates an Explained Alert Card, maps STIX/MITRE entities, and saves to `darkwatch.db`.